

HW5

2025-12-07

3230106230 刘仁钦

1. Yes. Noise in physical lines may cause errors in the transmitted data. There is a very small change that the error happens in both the destination IP and the checksum, and coincidentally the checksum can pass the verification. In this case, the packet will be delivered to the wrong destination.

2. $0xC2 = 194$
 $0x2F = 47$
 $0x15 = 21$
 $0x82 = 130$

The IP address in dotted decimal notation is 192.47.21.130.

3. (a). 135.46.63.10

Matches 135.46.60.0/22 (60 = 00111100, 63 = 00111111, the first 22 bits match).

Next hop: 159.48.0.1

(b). 135.46.57.14

Matches 135.46.56.0/22 (56 = 00111000, 57 = 00111001, the first 22 bits match).

Next hop: 211.90.0.1

(c). 135.46.52.2

Does not match 56.0/22 or 60.0/22.

Next hop: 220.20.0.1 (Default)

(d). 192.53.40.7

Matches 192.53.40.0/23.

Next hop: 192.188.0.1

(e). 192.53.56.7

Does not match 192.53.40.0/23 (40=00101000, 56=00111000, the 21st bit differs).

Next hop: 220.20.0.1 (Default)

4. Subnet mask 255.255.240.0 corresponds to /20.

Host bits = $32 - 20 = 12$

Maximum addresses = $2^{12} = 4096$.

Usable hosts = $4096 - 2 = 4094$

5. Original Packet: Data 900B + TCP Header 20B = 920B.

Total IP Length = 940B.

Link A-R1 (MTU 1010): No fragmentation needed.

Link R1-R2 (MTU 504): Fragmentation required.

- Max IP Payload = $504 - 20 = 484$.

Must be a multiple of 8 $\rightarrow$ 480B.

Fragment 1: Payload 480B. Total = 500B. Offset = 0. MF = 1

Fragment 2: Remaining Payload $920 - 480 = 440$ B.

Total = $440 + 20 = 460$ B. Offset = $\frac{480}{8} = 60$. MF = 0.

Link R2-B (MTU 500): Both fragments fit directly; no further fragmentation needed.

Link	Packet No.	Total length	MF	Fragment offset
A $\rightarrow$ R1	1	940	0	0
R1 $\rightarrow$ R2	1	500	1	0
R1 $\rightarrow$ R2	2	460	0	60
R2 $\rightarrow$ B	1	500	1	0
R2 $\rightarrow$ B	2	460	0	60

6.

Org No.	First IP	Last IP	net/mask
A	198.16.0.0	198.16.15.255	198.16.0.0/20
B	198.16.16.0	198.16.23.255	198.16.16.0/21
C	198.16.32.0	198.16.47.255	198.16.32.0/20
D	198.16.64.0	198.16.95.255	198.16.64.0/19

7. C

8. B

9. D